

Multiplying and Dividing in Scientific Notation*Question Bank - Additional 4.02*

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(3 \times 10^2)(3 \times 10^6)$

2. $\frac{6.3 \times 10^7}{2.1 \times 10^3}$

3. $(4 \times 10^5)(8 \times 10^6)$

4. $\frac{2.4 \times 10^4}{8 \times 10^7}$

5. $(2.5 \times 10^{-5})(6 \times 10^{-4})$

6. $\frac{9.9 \times 10^{-3}}{3.3 \times 10^{-8}}$

Real quantities.

7. The Amazon River discharges about 5.5×10^7 gallons of water every second.
- a. About how many gallons does it discharge in one hour, which is 3.6×10^3 seconds?
 - b. A second river discharges about 1.1×10^7 gallons per second. How many times larger is the Amazon's rate?
8. A factory produces about 6.4×10^5 bottles per day.
- a. How many bottles does it produce in 3.5×10^2 days?
 - b. A second factory produced 7.2×10^8 bottles in 1.2×10^3 days. How many bottles per day is that?

Working backwards.

9. Find the missing number in each equation.

a. $(4 \times 10^n)(2 \times 10^3) = 8 \times 10^{10}$

b. $\frac{7.2 \times 10^6}{n \times 10^2} = 2.4 \times 10^4$

c. $(2.5 \times 10^3)(8 \times 10^5) = \underline{\hspace{2cm}}$

Reading someone else's work.

10. Dana divides $\frac{4.8 \times 10^{11}}{1.2 \times 10^4}$ and writes 3.6×10^7 . Name her mistake and give the correct quotient.

11. Which is the value of $\frac{9.6 \times 10^{-3}}{3.2 \times 10^5}$?

- ☐ A. 3×10^{-8}
- ☐ B. 3×10^2
- ☐ C. 3×10^8
- ☐ D. 6.4×10^{-8}

12. Select ALL of the expressions whose value is 8×10^6 .

- ☐ A. $(4 \times 10^2)(2 \times 10^4)$
- ☐ B. $\frac{1.6 \times 10^{10}}{2 \times 10^3}$
- ☐ C. $(1.6 \times 10^3)(5 \times 10^3)$
- ☐ D. $\frac{4 \times 10^8}{5 \times 10^1}$
- ☐ E. $\frac{1.6 \times 10^{10}}{2 \times 10^4}$
- ☐ F. $(4 \times 10^2)(2 \times 10^8)$